Homework 3 - Aquaculture Analysis

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West Coast Aquaculture Analysis

In this project, we create a function that creates a map of suitable aquaculture area in Exclusive Economic Zones (EEZ) on the West Coast. The function takes in a specific species and its temperature and depth ranges and then ranks EEZ based on total suitable area (km^2) within each zone.

Load Data

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
library(terra)

sst  <- c(
    rast('data/average_annual_sst_2008.tif'),
    rast('data/average_annual_sst_2009.tif'),
    rast('data/average_annual_sst_2010.tif'),
    rast('data/average_annual_sst_2011.tif'),
    rast('data/average_annual_sst_2012.tif')
)

depth  <- rast("data/depth.tif")

eez <- st_read(here('data', 'wc_regions_clean.shp'))
```

```
Reading layer `wc_regions_clean' from data source
  `/Users/eriny/Documents/GitHub/eds223/EDS223-HW3/data/wc_regions_clean.shp'
```

```
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

Data Processing and Cleaning

Before making the function, we first prepare sea surface temperature and depth raster data for mapping. The function runs assuming this data processing has already happened.

```
# check crs are the same
if(st_crs(sst) != st_crs(depth)) warning("crs are not the same! fix!")
if(st_crs(sst) != st_crs(eez)) warning("crs are not the same! fix!")
```

```
# mean SST between 2008-2012
avg_sst <- mean(sst, na.rm = TRUE)
avg_sst <- avg_sst - 273.15
```

```
# resample depth raster to match extent of sst
#| results: hold

depth_resampled <- resample(depth, avg_sst, method = "bilinear")

cat("After resampling:\n")
```

```
After resampling:
```

```
cat("Resolutions match:", all(res(avg_sst) == res(depth_resampled)), "\n")
```

```
Resolutions match: TRUE
```

```
cat("Origins match: ", all(round(origin(avg_sst), 8) ==
                             round(origin(depth_resampled), 8)), "\n")
```

```
Origins match:  TRUE
```

```r
cat("Extents match: ", ext(avg_sst) == ext(depth_resampled), "\n")
```

```
Extents match:  TRUE
```

```r
# try stacking
depth_resampled <- project(depth_resampled, avg_sst) # ensure crs are actually the same

stack <- c(avg_sst, depth_resampled) # stack to check crs, resolution, extent are the same
```

Find Suitable Locations

We first want to run our analysis on oysters before generalizing the workflow to be applicable to any marine species. We find suitable cells on the map by selecting only cells that are suitable in both temperature and depth.

```r
# variables for min temp, max temp, min depth, max depth
oyster_temp_min <- 11
oyster_temp_max <- 30
oyster_depth_min <- -70
oyster_depth_max <- 0

# reclassify sst
mat <- matrix(c(oyster_temp_min, oyster_temp_max, 1,
-Inf, oyster_temp_min, 0,
oyster_temp_max, Inf, 0),
ncol=3, byrow=TRUE
)

avg_sst_class <- classify(avg_sst, mat)

# reclassify depth

d_mat <- matrix(c(oyster_depth_min, oyster_depth_max, 1,
-Inf, oyster_depth_min, 0,
oyster_depth_max, Inf, 0),
ncol = 3, byrow = TRUE)

depth_class <- classify(depth_resampled, d_mat)

# reclassify to identify sst and depth suitable raster
```

```r
suitable <- lapp(c(avg_sst_class, depth_class), fun = function(s, d) s * d)
names(suitable) <- "oyster suitability"
```

Determine Most Suitable EEZ

Total suitable area is determined by calculating area (km^2) per cell, and then multiplying that by each cell denoted '1', or suitable, in the previous code chunk. Total suitable area per zone is then calculated by summing the suitable areas in a zonal function.

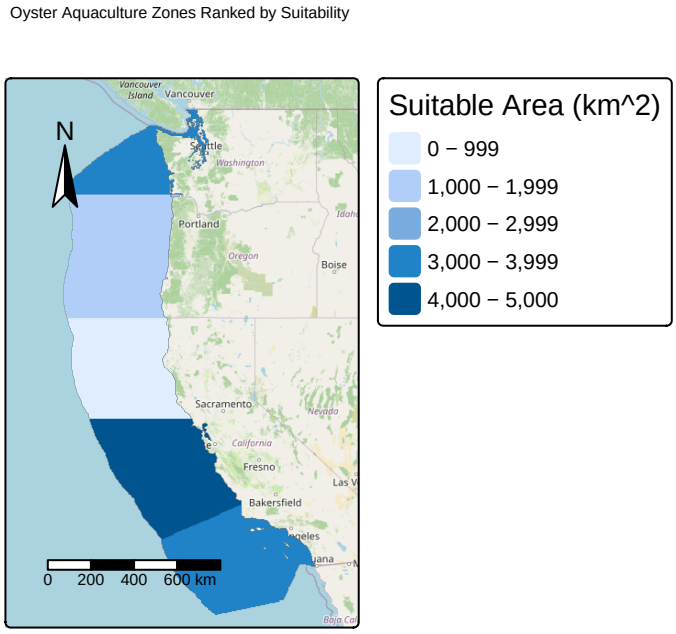
```r
# suitable cells are those denoted '1'
# rasterize eez
eez_rast <- as.factor(rasterize(eez, suitable, field = 'rgn_id'))

# calculate cell area
area <- cellSize(eez_rast, unit = 'km')

# calculate total suitable area per zone
suitable_area <- lapp(c(area, suitable), fun = function(a, s) a * s) # make mini multiply fu
eez_suitable_area <- zonal(suitable_area, eez_rast, fun = 'sum', na.rm = TRUE, as.raster = T
names(eez_suitable_area) <- 'Suitable Area (km^2)'
```

Now the data is ready to be mapped!

```r
# map
tm_shape(eez_suitable_area) +
  tm_raster() +
  tm_basemap('OpenStreetMap') +
  tm_title('Oyster Aquaculture Zones Ranked by Suitability') +
  tm_scalebar(position = c("left", "bottom")) +
  tm_compass(position = c("left", "top"))
```



It seems that the coast of Central California has the greatest area of suitable ocean for oyster aquaculture, at about 5,000 km^2. Next are Southern California and Washington, at about 3,000 km^2. And last are Oregon and Northern California, which have <2,000 km^2 of suitable area. These zones are not recommended for oyster aquaculture.

Making it a Function

The function is made using the code previously used to analyze oyster aquaculture suitability. It takes in a species name, minimum temperature, maximum temperature, minimum depth, and maximum depth. It then outputs a custom map with suitable area in West Coast aquaculture zones for the specified species.

```
map_suitable_eez <- function(tmin = 11, tmax = 30, dmin = -70, dmax = 0, species = 'Oyster')
  # reclassify sst
  mat <- matrix(c(tmin, tmax, 1,
  -Inf, tmin, 0,
  tmax, Inf, 0),
  ncol=3, byrow=TRUE
  )

  avg_sst_class <- classify(avg_sst, mat)
```

```
  # reclassify depth

  d_mat <- matrix(c(dmin, dmax, 1,
  -Inf, dmin, 0,
  dmax, Inf, 0),
  ncol = 3, byrow = TRUE)

  depth_class <- classify(depth_resampled, d_mat)

  # make sure reclassified sst and depth are the same crs
  if(st_crs(avg_sst_class) != st_crs(depth_class)) stop("crs are not the same! fix!")

  # reclassify to identify sst and depth suitable raster
  suitable <- lapp(c(avg_sst_class, depth_class), fun = function(s, d) s * d)

  # rasterize eez
  eez_rast <- as.factor(rasterize(eez, suitable, field = 'rgn_id'))

  # calculate cell area
  area <- cellSize(eez_rast, unit = 'km')

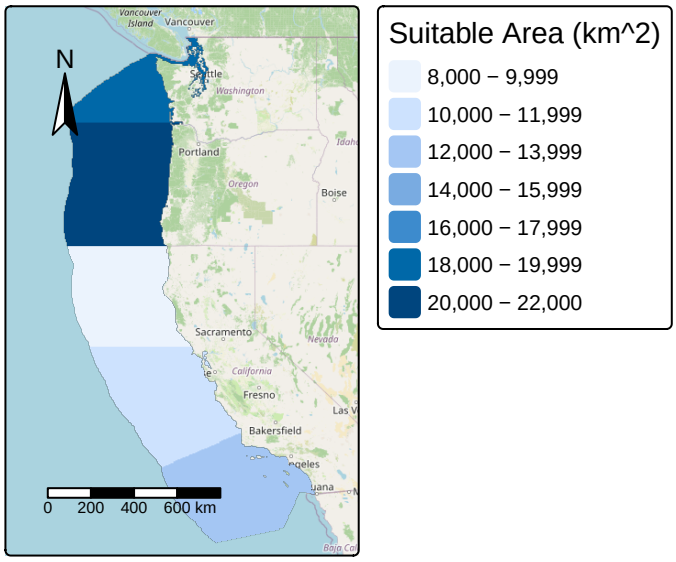
  # calculate total suitable area per zone
  suitable_area <- lapp(c(area, suitable), fun = function(a, s) a * s) # make mini multiply
  eez_suitable_area <- zonal(suitable_area, eez_rast, fun = 'sum', na.rm = TRUE, as.raster =
  names(eez_suitable_area) <- 'Suitable Area (km^2)'

  # map
  tm_shape(eez_suitable_area) +
    tm_raster() +
    tm_basemap('OpenStreetMap') +
    tm_title(paste0(species, ' Aquaculture Zones Ranked by Suitability')) +
    tm_scalebar(position = c("left", "bottom")) +
    tm_compass(position = c("left", "top"))
}
```

Let's try our function on a popular seafood item, Dungeness Crab!

```
map_suitable_eez(3, 19, -360, 0, 'Dungeness Crab')
```

Dungeness Crab Aquaculture Zones Ranked by Suitability



The most suitable area for Dungeness Crab is Oregon with about 20,000 km^2 of suitable area. Next is Washington, which also has considerable suitable area. California has less suitable area, around 8,000 to 15,000 km^2.